C-LiSA: A Hierarchical Hybrid Transformer Model Using Orthogonal Cross Attention for Satellite Image Cloud Segmentation

Subhajit Paul, Ashutosh Gupta, S. Manthira Moorthi, and Debajyoti Dhar

***Abstract*— Clouds in optical satellite images are a major concern since their presence hinders the ability to carry out accurate analysis as well as processing. Therefore, deriving accurate pixel-wise cloud masks is a key task in optical remote sensing. Several traditional as well as deep-learning algorithms have emerged to address this problem. However, the task of deriving accurate cloud masks from a variety of satellite images remains elusive due to the presence of confusing spectral signatures and changes in the properties of imaging sensors. In this article, we introduce a deep-learning model based on a hybrid transformer architecture for effective cloud mask generation named C-LiSA - cloud segmentation via the Lipschitz stable attention network. We propose two key attention mechanisms - dual orthogonal self-attention (DOSA) for handling confusing spectral signatures, and the hierarchical cross-channel attention (HC2A) model for effectively highlighting cloud-specific features during the cloud segmentation process. To validate the effectiveness of these mechanisms, we carry out theoretical and empirical Lipschitz stability analysis. We design the whole setup under an adversarial setting in the presence of Lovász-Softmax loss. We demonstrate both qualitative and quantitative outcomes for multiple satellite image datasets including Landsat-8, Sentinel-2, and Cartosat-2S. Our comparative study shows that the proposed model performs better compared to other state-of-the-art methods while providing better generalization across different datasets. We also showcase ablation studies to endorse our choices corresponding to different architectural elements and objective functions.**

***Index Terms*— Cloud segmentation, cross-attention, hybrid transformer, Lipschitz stability, orthogonal attention.**

I. Introduction

CLOUDS in satellite images are of immense interest to the meteorological community, but for optical remote sensing applications, cloud presence poses several challenges. Large cloud-covered data is difficult to process and analyze. Geophysical parameters such as radiance and brightness temperature derived from cloud-affected optical images do not match well with ground-truth values. Data processing steps such as ortho-rectification and image filtering also need to be aware of cloudy pixels to avoid outliers and artifacts in processed data. Clouds also affect satellite tasking required to cover an area of interest at a given time window. Lots of cloudy data is often acquired, transmitted, and stored in data processing systems essentially wasting valuable bandwidth and storage resources. Hence, a precise pixel-wise cloud mask generation model can be very useful to avoid artifacts and improve the overall usability of remote sensing data.

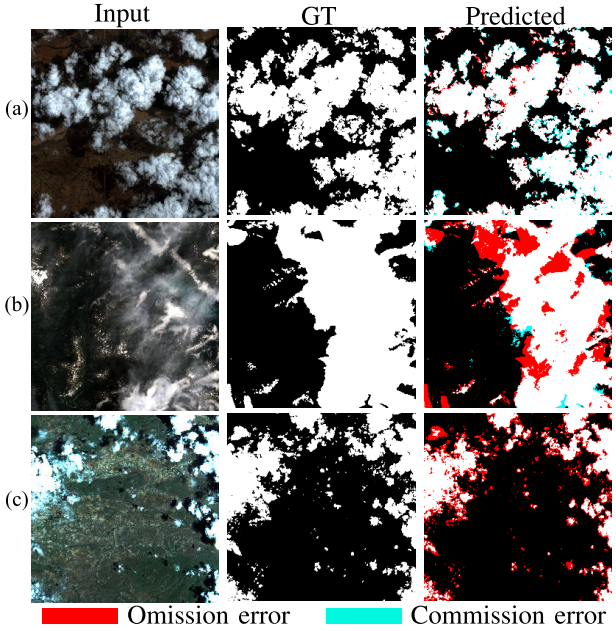


Fig. 1. Sample results for cloud mask generation for different satellites. (a) Cartosat-2S. (b) Landsat-8. (c) Sentinel-2A illustrates generalization and precise cloud mask generation capabilities sometimes even compared to GT in complex situations like haze in (b).

Traditional cloud mask generation methods utilize thresholding, spatial filtering, texture analysis, heuristics, and feature-wise classification. However, these methods suffer from several limitations since the clouds exhibit large variations in thickness, type, and scale which make it difficult to put them under a single "cloudy" class during the classification process. Also, the texture and color of ground features such as snow, ice, white sand, and so on often match that of clouds, further affecting the overall accuracy (OA) of these cloud masking techniques. Recently, deep-learning techniques have achieved impressive results for the cloud segmentation task, owing to their capability of better feature representation.

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The authors are with the Signal and Image Processing Area (SIPA), Space Applications Center (SAC), ISRO, Ahmedabad 380015, India (e-mail: subhajitpaul@sac.isro.gov.in).

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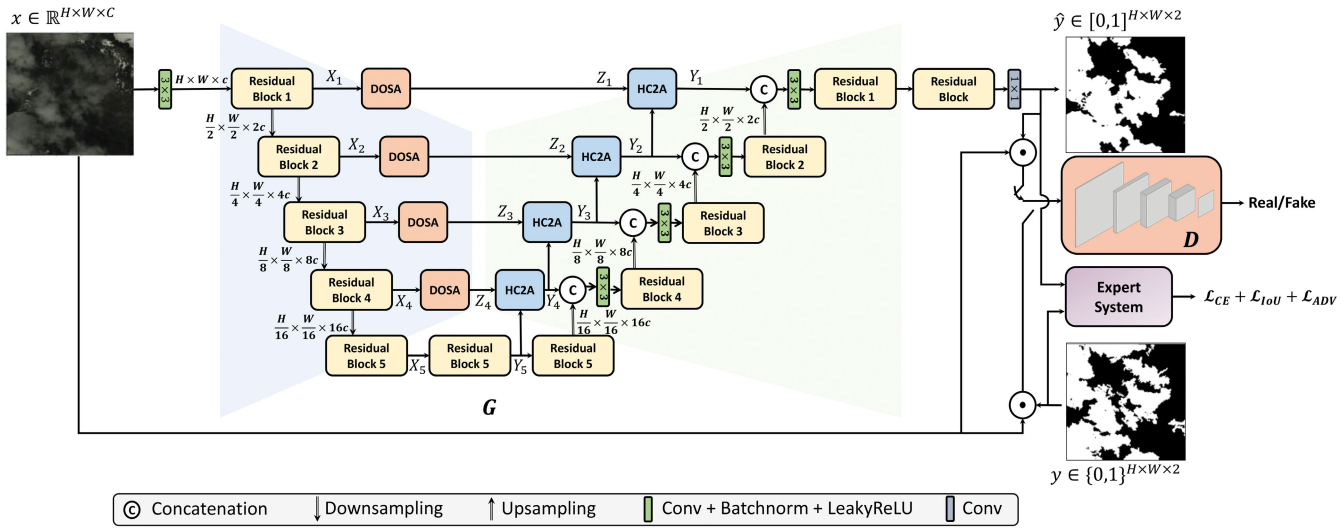


Fig. 2. Overview of our generative framework with the proposed architecture of hierarchical hybrid transformer. The generator G features DOSA and HC2A, two novel attention modules in every skip connection to hierarchically transfer the dense features to shallow layers for efficient segmentation.

However, most of these techniques fail to perform impressively under the presence of confusing spectral signatures, different spatial and spectral resolutions, dynamic ranges, noise levels, and spectral band definitions. A good cloud masking model should work efficiently in the presence of confusing ground features as well as have better generalization capability to act on images with different cloud types, and spatial and spectral resolution. In this study, we propose a novel and robust cloud segmentation framework for intricate cases that also showcases better generalization for cross-domain inference with minimal transfer learning. Fig. 1 illustrates such performance of our model for various satellite images, showcasing the capability of generalization, sometimes even better compared to ground truth (GT) in complex scenarios like haze [Fig. 1(b)]. Our key contributions can be summarized as follows.

1) We propose a novel hybrid transformer architecture for cloud segmentation, namely cloud segmentation via the Lipschitz stable attention (C-LiSA), based on dual orthogonal self-attention (DOSA) and HC2A.
2) We provide theoretical and empirical justification for the Lipschitz stability of our proposed attention modules and compare them with other popular attention models.
3) We devise an objective function based on Lovász-Softmax loss in an adversarial setup that promotes cloud masks with refined boundaries.
4) We perform experiments on four different satellite image datasets with different band combinations and perform ablation studies to show the efficacy of our model.

The rest of the article is organized as follows. In Section II, we discuss relevant prior works. We discuss detailed methodology including overall architecture and loss formulation in Section III. Dataset selection and implementation details are provided in Section IV. Detailed experimental results and comparative analysis are carried out in Section V followed by an ablation study in Section VI and complexity analysis in Section VII. Finally, we draw our concluding remarks in Section VIII.

II. Related Work

Cloud detection from satellite images has been an active research area in the domain of remote sensing. Traditionally, cloud mask was generated from satellite images via thresholding [1] and region or contour-based image segmentation [2]. However, these methods suffer in the presence of variations in imaging conditions and complex cloud patterns. To resolve these, one of the first effective cloud detection methods was Fmask [3], [4]. Fmask uses spectral information and correlation between different bands of multispectral (MX) imagery to segment pixels into cloudy, noncloudy, and shadow classes. Although this works preferably in most situations, their thresholding criteria based on spectral analysis between different bands is affected by factors like atmospheric conditions, noise, and the availability of specific spectral bands.

Following the rise of deep-learning techniques, notable convolutional neural networks (CNNs) [5], [6] have been developed for semantic segmentation tasks like cloud mask generation. Francis et al. [7] introduced CloudFCN as one of the leading-edge models for pixel-level cloud segmentation. They developed this model based on U-Net [8] with the addition of inception modules between skip connections. Yang et al. [9] proposed a CNN-based cloud segmentation module, namely CDNetV1, which has a built-in boundary refinement module for thumbnail images of the ZY-3 satellite. Later, they proposed an updated network CDNetV2 [10], equipped with a new feature fusion and information guidance module for precise cloud mask extraction. This was designed specifically to work under complex conditions like scenarios of snow-cloud coexistence. Jeppesen et al. [11] introduced the RS-Net model that utilizes U-Net for cloud segmentation. They trained their model with a manually generated cloud mask as well as a mask generated using the Fmask algorithm on Landsat images. Their model, trained with outputs from Fmask, outperforms the outcomes obtained directly from Fmask. In recent work, Jiao et al. [12] proposed a Refined-UNet-V4 (RefUNetV4) which utilizes U-Net to coarsely locate

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cloudy regions followed by an upgraded conditional random field (CRF) for refining edges.

After the introduction of a transformer for vision tasks [13], multiple transformer-based architectures have been derived for image segmentation [14], [15]. These models excel at learning rich and abstract feature representations from the input. For all image regions, they capture relations between spatially distant but semantically related areas using self-attention. However, as transformer models majorly excel in global feature extraction, they cannot exact local dense features compared to CNN. To leverage the benefit of both the CNN and transformer model, several hybrid transformer models have been proposed recently. Gong et al. [16] and Liu et al. [17] have proposed such hybrid-transformer models to show their efficacy in cloud segmentation applications. However, due to the adverse stability of the attention modules, these methods lack generalization capability and are prone to different adversarial phenomena. In this article, we propose a novel hybrid transformer model based on a Lipschitz stable attention module consisting of a DOSA and a hierarchical channel cross-attention (HC2A) module. The main reason to ensure Lipschitz stability is to make our model robust against different adversarial attacks as discussed in [18]. We model our framework under an adversarial setup to reduce the possibility of overfitting during model training [19]. In Section III, we describe our overall methodology.

III. Methodology

Fig. 2 demonstrates the overview of our proposed framework. The generator model G operates on multispectral (MX) satellite image patches $\mathbf{x} \in \mathbb{R}^{H\times W\times C}$, where C corresponds to the number of channels in the input patch with height H and width W. Let $\mathbf{x} \sim \mathbb{P}_X$, where $\mathbb{P}_X$ is the joint source distribution depending on C. Let $\mathbf{y} \sim \mathbb{P}_Y$ be the target distribution with $\mathbf{y} \in \{0,1\}^{H\times W\times m}$, where m is the total number of classes specified for segmentation and let $\hat{\mathbf{y}} \sim \mathbb{P}_{G_\theta}$, where $\hat{\mathbf{y}} = G_\theta(\mathbf{x})$ with $\hat{\mathbf{y}} \in [0,1]^{H\times W\times m}$ and $\mathbb{P}_{G_\theta}$ be the generator distribution parameterized by $\theta \in \Theta$. The discriminator D classifies GT data and generated data as real and fake, respectively.

A. Network Architecture

We propose a novel hierarchical hybrid transformer architecture based on adversarial learning, as shown in Fig. 2. We use the discriminator of PatchGAN [20] as D, whereas the generator model G comprises an encoding part and a decoding part similar to the U-Net model. The encoding part extracts high-resolution features and dense contextual information from cloudy input images while the decoding part works on these features to generate the cloud segmentation map. Unlike traditional U-Net, both of these components are incorporated with residual blocks along with upsampling and downsampling operations as shown in Fig. 2. These residual blocks consist of CNN operations with multiple residual connections as shown in Fig. 3(a) which ensures efficient context flow throughout the network. Similar to U-Net, we use skip connections to assimilate the encoding and decoding feature maps to assist the decoder in effective reconstruction while avoiding the vanishing gradient problem. However, instead of direct skip connections, we augment it with our attention modules DOSA and HC2A, inspired by the multi-Dconv head transposed attention (MDTA) of Restormer [21] as shown in Fig. 2. Their detailed architectures are illustrated in Fig. 3. As we demonstrate later, this helps to suppress activations in irrelevant regions by reducing the number of redundant features, especially for initial layers. We now present the details of these attention modules.

1) Dual Orthogonal Self-Attention: The activations in initial layers tend to distribute equal weights for every feature ultimately resulting in poor segmented outcomes. In this work, to enable accurate cloud mask generation, we design DOSA to capture long-range spectral and structural information from the feature maps. More specifically, it enables the effective extraction of distinguishable and correlated features among different signatures in the global context and updates the weights according to the significance of different semantic features. DOSA is designed to preserve high-resolution semantics during computing attention in pixel-wise regression compared to other self-attention modules available in the literature. For leveraging important structural and spectral features related to clouds in both channel and spatial direction, it simultaneously computes the attentions in two orthogonal branches, spatial-only-attention ($A^{\text{sp}}(\cdot)$) and channel-only attention ($A^{\text{ch}}(\cdot)$). Each branch is inspired from [22] and designed from the concept of polarizing filter in photography by completely collapsing feature maps in one direction while preserving high resolution in its orthogonal direction and also high dynamic range (HDR) operation by using Softmax and Sigmoid normalization for the smallest feature map to recover crucial details. The intrinsic architectural details of DOSA are demonstrated in Fig. 3.

In both attention branches, the query and key matrix are estimated as orthogonal to each other. In channel-only-attention branch, for an input $\mathbf{X} \in \mathbb{R}^{H\times W\times C}$, the query and key are computed as $\mathbf{Q_C} \in \mathbb{R}^{HW\times C}$ and $\mathbf{K_C} \in \mathbb{R}^{1\times HW}$, respectively.

For input $\mathbf{X}$, the channel only attention is estimated such that $\mathbf{A^{ch}}(\mathbf{X}) \in \mathbb{R}^{1\times 1\times C}$ and it is computed as

$$\mathbf{A^{ch}}(\mathbf{X}) = \sigma\left(F_{\text{RS}}\left[\mathbf{W_C^q} * \mathbf{X}\right] \times \phi\left(F_{\text{RS}}\left[\mathbf{W_C^k} * \mathbf{X}\right]\right)\right) \quad (1)$$

where $\mathbf{W_C^q}$ and $\mathbf{W_C^K}$ are 3×3 convolution operations with the corresponding weight matrix of query and key with $*$ being the convolution operation, F_{RS} is reshape operator, and $\sigma(\cdot)$ and $\phi(\cdot)$ are Sigmoid and Softmax functions, respectively. The output of the channel-only-attention is estimated as follows.

Similarly, for the spatial-only-attention, the corresponding query ($\mathbf{Q_S}$) and key ($\mathbf{K_S}$) are estimated such that $\mathbf{Q_S} \in \mathbb{R}^{HW\times C}$ and $\mathbf{K_S} \in \mathbb{R}^{1\times C}$, and the related attention $\mathbf{A^{sp}}(\mathbf{X}) \in \mathbb{R}^{H\times W\times 1}$ is approximated as

$$\mathbf{A^{sp}}(\mathbf{X}) = \sigma\left(F_{\text{RS}}\left[\phi\left(F_{\text{RS}}\left[F_{\text{GP}}\left(\mathbf{W_S^k} * \mathbf{X}\right)\right]\right) \times F_{\text{RS}}\left[\mathbf{W_S^q} * \mathbf{X}\right]\right]\right) \quad (2)$$

where $F_{\text{GP}}(\cdot)$ is a global pooling operator $F_{\text{GP}}(\mathbf{X}) = (1/(H\times W))\sum_{i=1}^{H}\sum_{i=1}^{W} X(:,i,j)$, $\mathbf{W_S^q}$, $\mathbf{W_S^k}$ are 3×3 convolution operations with the corresponding weight matrix of query, key during estimating spatial attention. The end product

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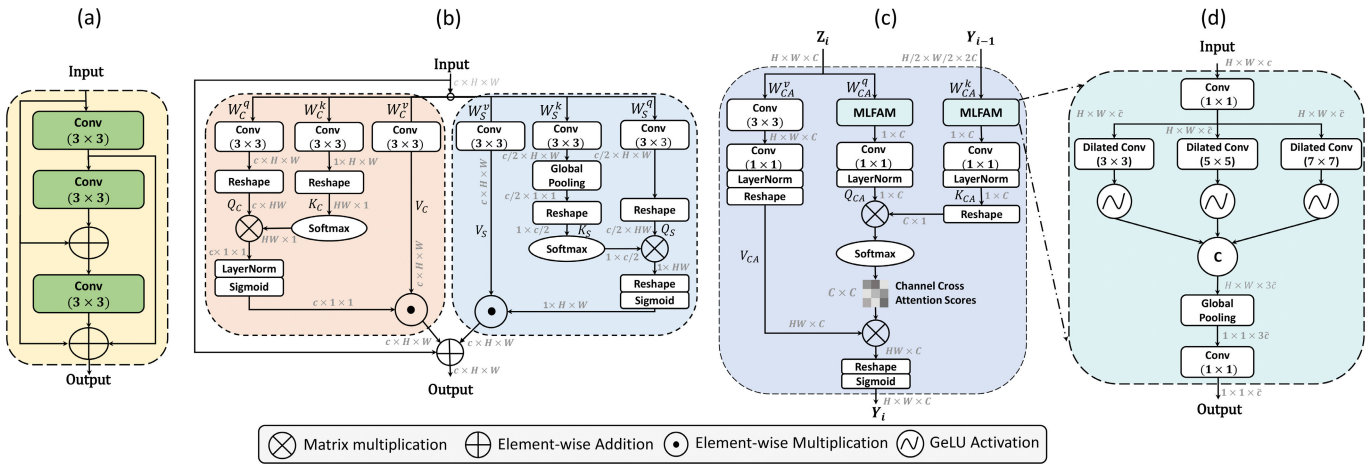


Fig. 3. Overview of proposed attention modules and their components. (a) Residual block. (b) DOSA. (c) HC2A. (d) LFAM.

of DOSA can be conveyed with a parallel layout of the outputs of the above two branches as shown in the following equation:

$$\mathbf{Z} = \mathbf{X} + \mathbf{A}^{\text{ch}}(\mathbf{X}) \odot \left(\mathbf{W}_{\mathbf{C}}^{\mathbf{v}} * \mathbf{X}\right) + \mathbf{A}^{\text{sp}}(\mathbf{X}) \odot \left(\mathbf{W}_{\mathbf{S}}^{\mathbf{v}} * \mathbf{X}\right) \quad (3)$$

where $\mathbf{Z} = \text{DOSA}(\mathbf{X})$, $\mathbf{W}_{\mathbf{C}}^{\mathbf{v}}$, and $\mathbf{W}_{\mathbf{S}}^{\mathbf{v}}$ are 3×3 convolution kernels corresponding to value matrix $\mathbf{V_C}$ and $\mathbf{V_S}$ for channel and spatial attention branch, respectively and $\odot$ denotes element-wise multiplication.

Another major motivation behind the design of DOSA is to provide better sensitivity in terms of Lipschitz constant. As shown in [23], the L_2-Lipschitz constant of the self-attention mechanism in the traditional transformer model is bounded by the variance of the input, resulting in larger sensitivity. In this context, we propose Theorem 1 to derive the approximate Lipschitz bound of the DOSA module to visualize its sensitivity with respect to any given input.

Theorem 1: *Let m and M be the minimum and maximum of input feature $\mathbf{X} \in \mathbb{R}^{H\times W\times C}$. Let $\omega = e^{(2\pi i/HW)}$ and W^q be the convolution kernel for the query matrix in DOSA. Let Γ be the difference between the learned kernel and its initialization and given by $\Gamma = W^q - W_0^q$. Let U be a complex matrix such that $U_{ij} = \omega^{ij}$. If $\varepsilon^q = (1/9)(U^T \Gamma U)_{0,0}$, then the upper bound on the magnitude of diagonal elements of Jacobian of DOSA is estimated as*

$$||J_{i,i}^{SA}||_2 \leq 1 + \frac{1}{2}\left(1 + 9\varepsilon_i^v\right)\left(1 + 9\varepsilon_i^q\right) \times \left(B + \sqrt{2}(1 + 9\varepsilon_i^k)B^3\right) + 2\left(1 + 9\varepsilon_i^v\right) \quad (4)$$

where, $B = (HW)^{1/2}\max(||m||, ||M||)$, ε_i^v and ε_i^k are corresponding to value and key matrix in DOSA.

Proof: This proof is inspired by [23] and an immediate application of operator norm for convolution kernels [18] which says 2-D convolution operations can be represented as matrix multiplication with corresponding operator matrix, that is, $W_f * x \overset{\text{def}}{=} \text{op}(W_f)x$. Therefore, the output of the channel attention branch using (1) can be rewritten as

$$\begin{aligned}\mathbf{Z}^{\text{ch}} &= \mathbf{A}^{\text{ch}}(\mathbf{X}) \odot \left(\text{op}\left(\mathbf{W}_{\mathbf{C}}^{\mathbf{v}}\right)\mathbf{X}\right) \\ &= \sigma\left(F_{\text{RS}}\left[\text{op}\left(\mathbf{W}_{\mathbf{C}}^{\mathbf{q}}\right)\mathbf{X}\right] \times \phi\left(F_{\text{RS}}\left[\text{op}\left(\mathbf{W}_{\mathbf{C}}^{\mathbf{k}}\right)\mathbf{X}\right]\right)\right) \\ &\quad \odot \left(\text{op}\left(\mathbf{W}_{\mathbf{C}}^{\mathbf{v}}\right)\mathbf{X}\right). \end{aligned} \quad (5)$$

To estimate the magnitude of Jacobian $\mathbf{Z}^{\text{ch}}$, the reshape operator $F_{\text{RS}}(\cdot)$ can be ignored, and hence, we can write

$$\begin{aligned}\left\|J_{i,i}^{\text{ch}}\right\| &= \left\|\frac{\partial \mathbf{Z}_{\mathbf{i}}^{\mathbf{ch}}}{\partial \mathbf{X}_i}\right\| \\ &= \Bigg\|\left(\text{diag}(\sigma(\boldsymbol{\alpha_1})) \odot \text{op}\left(\mathbf{W}_{\mathbf{C_i}}^{\mathbf{v}}\right)\right) \\ &\quad + \left(\text{diag}\left(\text{op}\left(\mathbf{W}_{\mathbf{C_i}}^{\mathbf{v}}\right)\mathbf{X}_i \odot \sigma(\boldsymbol{\alpha_1})(1 - \sigma(\boldsymbol{\alpha_1}))\right)\right) \\ &\quad \times \Bigg(\text{diag}\Bigg(\text{op}\left(\mathbf{W}_{\mathbf{C_i}}^{\mathbf{q}}\right).\phi(\beta_1) \\ &\quad + \text{op}\left(\mathbf{W}_{\mathbf{C_i}}^{\mathbf{q}}\right)\mathbf{X}_i.\frac{\partial}{\partial \mathbf{X}_i}\phi(\beta_1)\Bigg)\Bigg)\Bigg\| \end{aligned} \quad (6)$$

where $\alpha_1 = \text{op}(\mathbf{W}_{\mathbf{C}}^{\mathbf{q}})\mathbf{X} \times \phi(\text{op}(\mathbf{W}_{\mathbf{C}}^{\mathbf{k}})\mathbf{X})$ and $\beta_1 = \text{op}(\mathbf{W}_{\mathbf{C}}^{\mathbf{k}})\mathbf{X}$. Now using triangular and Cauchy–Schwarz inequality to (6), we will get

$$\begin{aligned}\left\|J_{i,i}^{\text{ch}}\right\| &\leq \left\|\left(\text{diag}(\sigma(\boldsymbol{\alpha_1})) \odot \text{op}\left(\mathbf{W}_{\mathbf{C_i}}^{\mathbf{v}}\right)\right)\right\| \\ &\quad + \left\|\left(\text{diag}\left(\text{op}\left(\mathbf{W}_{\mathbf{C_i}}^{\mathbf{v}}\right)\mathbf{X}_i \odot \sigma(\boldsymbol{\alpha_1})(1 - \sigma(\boldsymbol{\alpha_1}))\right)\right)\right\| \\ &\quad \times \Bigg(\left\|\text{diag}\left(\text{op}\left(\mathbf{W}_{\mathbf{C_i}}^{\mathbf{q}}\right).\phi(\beta_1)\right)\right\| \\ &\quad + \left\|\text{op}\left(\mathbf{W}_{\mathbf{C_i}}^{\mathbf{q}}\right)\mathbf{X}_i.\frac{\partial}{\partial \mathbf{X}_i}\phi(\beta_1)\right\|\Bigg). \end{aligned} \quad (7)$$

From [24], we know for any two matrix $\mathbf{A} \in \mathbb{R}^{d_1\times n}$ and $\mathbf{B} \in \mathbb{R}^{d_2\times n}$ dependent on X, the following inequality holds:

$$\left\|\mathbf{A}\frac{\partial}{\partial \mathbf{X}}\text{softmax}(\mathbf{B})\right\|_2 \leq \sqrt{2}||\mathbf{A}||_{(2,\infty)}||\mathbf{B}||_{(\infty,2)} \quad (8)$$

where $||\mathbf{B}||_{(\infty,2)} = \max_i(\sum_j B_{i,j}^2)^{1/2}$ [24]. Considering the norm in (7) as $L2$, and taking the operator norm for a $t \times t \times 1$ shaped kernel W_f as $||\text{op}(W_f)||_2 = 1 + \varepsilon t^2$, the Euclidean Lipschitz constant from (7) is estimated as

$$||J_{i,i}^{\text{ch}}||_2 \leq \frac{1}{4}\left(1 + 9\varepsilon_i^{v^C}\right)\left(1 + 9\varepsilon_i^{q^C}\right) \times \left(||\mathbf{X}_i||_2 + \sqrt{2}\left(1 + 9\varepsilon_i^{k^C}\right)||\mathbf{X}_i||_2^3\right) + \left(1 + 9\varepsilon_i^{v^C}\right). \quad (9)$$

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Similarly for the spatial attention branch, the Euclidean norm of the Jacobian from (2) is estimated as

$$||J_{i,i}^{\text{sp}}||_2 \leq \frac{1}{4}\left(1+9\varepsilon_i^{v^S}\right)\left(1+9\varepsilon_i^{q^S}\right) \times \left(||\mathbf{X}_i||_2+\sqrt{2}\left(1+9\varepsilon_i^{k^S}\right)||\mathbf{X}_i||_2^3\right)+\left(1+9\varepsilon_i^{v^S}\right). \tag{10}$$

For simplicity, we assume the convolutional kernels regarding key, query, and value matrix are identical for both channel and spatial attention. Then, taking $||X||_2 \leq (HW)^{1/2}\max(||m||, ||M||)$, the final Euclidean–Lipschitz constant of DOSA can be approximated using (3)

$$\left\|J_{i,i}^{\text{SA}}\right\|_2 = \left\|\frac{\partial \mathbf{Z_i}}{\partial \mathbf{X}_i}\right\|_2 \leq 1+\frac{1}{2}\left(1+9\varepsilon_i^{v}\right)\left(1+9\varepsilon_i^{q}\right) \times \left(B+\sqrt{2}\left(1+9\varepsilon_i^{k}\right)B^3\right)+2\left(1+9\varepsilon_i^{v}\right) \tag{11}$$

which completes the proof. □

Theorem 1 shows, unlike traditional self-attention, the Euclidean–Lipschitz constant of DOSA is upper bounded by the cube of the absolute maximum of input. Therefore, its performance is relatively stable for $[-1, 1]$ normalized input. From the perspective of our designed model, this is crucial as almost all input features to DOSA are $[-1, 1]$ bounded because of batchnorm and layernorm utilization. The output of DOSA is passed through the HC2A block, which we discuss next.

2) Hierarchical Cross-Channel Attention: Although DOSA ensures to update of the skip connection activation maps with respect to extracted long-range structural and spectral information, it does not guarantee to enhance the poor feature representations in the initial layers in terms of semantic richness, especially in scenarios like snow-cloud coexistence. However, the feature maps in the deeper network comprise dense semantic information that can be utilized to update the initial feature maps. This results in more accurate and generalized cloud mask extraction as well as enabling the potential to distinguish between confusing features like snow and cloud more effectively. HC2A module is designed based on this perception and its key aspect is to actively suppress activations in irrelevant regions, reducing the number of redundant features brought across for similar spectral signatures like snow and sand and highlighting only cloudy areas. HC2A module acts in a hierarchical manner, in which at particular downsampling level n, it generates the enhanced feature activations as $\mathbf{Y_n} = \text{HC2A}(\mathbf{Z_n}, \mathbf{Y_{n-1}})$ as shown in Fig. 2. The detailed architectural details of the HC2A module are illustrated in Fig. 3(c).

The inputs to HC2A, $\mathbf{Y_{n-1}}$ and $\mathbf{Z_n}$ are spatially different in shape. This leads to certain downsampling or upsampling operations to accumulate these features. However, these operations may blur out the dense semantic information which is mostly preserved in channel direction only at deeper levels. Hence, we propose Laplacian feature aggregation modules (LFAMs) for effective estimation of cross-attention weights by being independent of any spatial feature manipulation. HC2A consists of two LFAMs, which is a Laplacian pyramid structure with 3×3 convolution operations with dilation rates 3, 5, and 7 followed by GeLU activation functions. We combine these features and apply a 3×3 convolution as shown in Fig. 3(d). By employing LFAM, we ensure efficient feature computation and the generation of sparse channel activation maps by eliminating noisy spectral information in latent space.

As shown in Fig. 3(c), at particular downsampling level n, the query matrix $\mathbf{Q_{CA}}$ is computed by passing input $\mathbf{Z_n} \in \mathbb{R}^{H\times W\times C}$ through an LFAM block such that $\mathbf{Q_{CA}} \in \mathbb{R}^{1\times C}$. Similarly, the key matrix $\mathbf{K_{CA}}$ is estimated for input $\mathbf{Y_{n-1}} \in \mathbb{R}^{H/2\times W/2\times 2C}$ such that $\mathbf{K_{CA}} \in \mathbb{R}^{1\times C}$. Therefore, the output of the cross-attention block is formulated as in the following equation:

$$\mathbf{Y_n} = \sigma\left(F_{\text{RS}}\left[F_{\text{RS}}\left[\mathbf{W_{CA}^v} * \mathbf{Z_n}\right] \times \phi(\text{LFAM}(\mathbf{Z_n}) \times \text{LFAM}(\mathbf{Y_{n-1}}))\right]\right) \tag{12}$$

where $\mathbf{W_{CA}^v}$ is 3×3 is the convolution kernel to estimate the value matrix $\mathbf{V_{CA}}$ from input $\mathbf{Z_n}$. Similar to Theorem 1, we approximate the Lipschitz constant of HC2A and investigate its effect on the sensitivity of the overall attention module. In this context, we propose Theorem 2.

Theorem 2: *Using the definition of operator norm in Theorem 1, the bound on the magnitude of diagonal elements of Jacobian of HC2A can be given by*

$$\left\|J_{i,i}^{CA}\right\| \leq \frac{1}{2\sqrt{2}}\left(1+9\varepsilon_{i,n}^{v^{CA}}\right)\left(\frac{1}{\sqrt{2}}+\left(1+9\varepsilon_{i,n}^{q^{CA}}\right) \times \left(1+9\varepsilon_{i,n}^{k^{CA}}\right)||\mathbf{Z_{i,n}}||^2.||\mathbf{Y_{i,n-1}}||\right) \tag{13}$$

where $\varepsilon_n^{q^{CA}}$, $\varepsilon_n^{k^{CA}}$, and $\varepsilon_n^{v^{CA}}$ correspond to query, key, and value matrix in HC2A, respectively. Then, the overall Jacobian of DOSA + HC2A is bounded $\mathcal{O}(B^4+B^6)$ with $B = (HW)^{1/2}\max(||m||, ||M||)$ for an input $\mathbf{X_n} \in \mathbb{R}^{H\times W\times C}$.

Proof: For simplicity, we assume the operation of LFAM similar to 3×3 convolution operations with $\mathbf{W_{CA}^q}$ and $\mathbf{W_{CA}^k}$ kernels for key and query matrix as shown in Fig. 3(a) and write the expression for cross attention as

$$\mathbf{Y_n} = \sigma\left(F_{\text{RS}}\left[F_{\text{RS}}\left[\text{op}(\mathbf{W_{CA}^v})\mathbf{Z_n}\right] \times \phi\left(\text{op}\left(\mathbf{W_{CA}^q}\right)\mathbf{Z_n} \times \text{op}\left(\mathbf{W_{CA}^k}\right)\mathbf{Y_{n-1}}\right)\right]\right). \tag{14}$$

Then, using steps similar to **Thereom 1**, we can estimate the bound on the magnitude of diagonal elements of Jacobian of HC2A as defined in (13).

As almost all the feature maps are bounded in $[-1, 1]$ due to usage of layer-norms, and $\mathbf{Z_n} = \text{DOSA}(\mathbf{X_n})$, we can assume $\|\mathbf{Z_n}\|_2 \leq B$ and $||\mathbf{Y_{n-1}}||_2 \leq (B/2)$. Therefore, we can simply rewrite (13) as $||J_{i,i}^{\text{CA}}|| = ||(\partial \mathbf{Y_i}/\partial \mathbf{Z_i})|| \leq \mathcal{O}(B^3)$. Similarly, we can simplify (11) to write $||J_{i,i}^{\text{SA}}|| = ||(\partial \mathbf{Z_i}/\partial \mathbf{X_i})|| \leq \mathcal{O}(B+B^3)$. Then, to approximate the magnitudes on diagonals of Jacobian for the overall DOSA + HC2A module, we apply the chain rule and Cauchy–Schwarz inequality to obtain the final bound as shown below

$$||J_{i,i}||_2 = \left\|\frac{\partial \mathbf{Y_{i,n}}}{\partial \mathbf{X_{i,n}}}\right\|_2 \leq \left\|\frac{\partial \mathbf{Y_{i,n}}}{\partial \mathbf{Z_{i,n}}}\right\|_2 \left\|\frac{\partial \mathbf{Z_{i,n}}}{\partial \mathbf{X_{i,n}}}\right\|_2 \leq \mathcal{O}(B^4+B^6) \tag{15}$$

which completes the proof. □

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Theorem 2 shows that besides providing dense features to accurately generate segmentation map, HC2A improves the overall Lipschitz sensitivity which makes it stable for normalized $[-1, 1]$ bounded input, improving cloud mask generation even in the presence of any adversary such as noise. Later, we empirically justify this claim in the ablation section.

B. Network Objective Definition

The overall framework is optimized using a set of loss functions along with the minimax objective function of conditional GAN [20]. In the following sections, we define the objective functions to train generator G and discriminator D.

1) Discriminator Objective Function: Let the discriminator model D be parameterized by $\psi \in \Psi$. D_ψ takes generated mask $\hat{\mathbf{y}}$ and true label $\mathbf{y}$, both conditioned on input $\mathbf{x}$ and classifies the corresponding patch as real or fake, respectively. Thus, the objective function of D_ψ is defined as

$$\min_{\psi} -\mathbb{E}_{\mathbf{x}\sim\mathbb{P}_X,\mathbf{y}\sim\mathbb{P}_Y}[\log(D_\psi(\mathbf{x},\mathbf{y}))] - \mathbb{E}_{\mathbf{x}\sim\mathbb{P}_X,\hat{\mathbf{y}}\sim\mathbb{P}_{\mathbf{G}_\theta}}[\log(1-D_\psi(\mathbf{x},\hat{\mathbf{y}}))]. \quad (16)$$

2) Generator Objective Function: The generator model G, parameterized with $\theta \in \Theta$, yields segmentation map $\hat{\mathbf{y}}$. The prime objective of G is to generate $\hat{\mathbf{y}}$ closely related to the true label $\mathbf{y}$ while maintaining underlying statistical attributes close to the real manifold. For this purpose, G_θ is trained with a combination of Focal loss and Lovász-Softmax loss. To match statistical measures between y and $\hat{y}$, it is also regularized with the adversarial loss. Next, we discuss these losses in detail.

a) Focal loss: The main role of Focal loss is similar to cross-entropy (CE) loss which provides a measure of dissimilarities between predicted probabilities and GT labels. The advantage of focal loss over CE loss is its ability to focus on examples that are difficult to classify like thin clouds while reducing the influence of easily classified ones. This is done by tuning a focusing parameter γ in the following loss function:

$$\mathscr{L}_{\text{CE}} = \mathbb{E}_{\mathbf{y}\sim\mathbb{P}_Y,\hat{\mathbf{y}}\sim\mathbb{P}_{\mathbf{G}_\theta}} - w(1-\hat{\mathbf{y}})^\gamma \log(\hat{\mathbf{y}}) \quad (17)$$

where $w \in \mathbb{R}^m$ are weighted for m number of classes to handle class imbalance during loss computation. Focal loss similar to CE is very sensitive to outliers or mislabeled classes and is unable to accurately delineate object boundaries resulting in poor generalization. To overcome these limitations, leveraged the usage of Lovász-Softmax loss to train G.

b) Lovász-softmax loss: Jaccard index, also referred to as the intersection-over-union (IoU) score, is one of the key performance measures in semantic segmentation suggesting the amount of intersection between a set of true labels and generated labels. Hence, it is often used as a loss function for not only boundary enhancement but also the precise location of the generated segmentation map. For GT label $\mathbf{y}$ and generated label $\hat{\mathbf{y}}$, the Jaccard loss $(\Delta_{\mathcal{J}_c})$ for a class $\mathrm{c} \in \mathcal{C}$ can be written as a set of mispredictions $(\mathbf{M}_c)$ as shown below

$$\mathbf{M}_c(\mathbf{y},\hat{\mathbf{y}}) = \{\mathbf{y}=\mathrm{c}, \hat{\mathbf{y}}\neq\mathrm{c}\} \cap \{\mathbf{y}\neq\mathrm{c}, \hat{\mathbf{y}}=\mathrm{c}\}$$
$$\Delta_{\mathcal{J}_c} : \mathbf{M}_c \in \{0,1\}^p \mapsto \frac{|\mathbf{M}_c|}{|\{\mathbf{y}=\mathrm{c}\}\cap\mathbf{M}_c|} \quad (18)$$

where p is the number of pixels in the minibatch. However, as $(\Delta_{\mathcal{J}_c})$ is a set function, its optimization is intractable due to its nonsmooth properties. To circumvent this, we apply Lovász-Softmax loss $(\mathscr{L}_{\mathcal{J}})$ [25], which is a convex surrogate of the Jaccard loss

$$\mathscr{L}_{\mathcal{J}} = \mathbb{E}_{\mathbf{y}\sim\mathbb{P}_Y,\hat{\mathbf{y}}\sim\mathbb{P}_{\mathbf{G}_\theta}} \frac{1}{|\mathcal{C}|}\sum_{\mathrm{c}\in\mathcal{C}} \overline{\Delta_{\mathcal{J}_c}}(\phi(\mathbf{M}_c(\mathbf{y},\hat{\mathbf{y}})))$$
$$\text{where } \overline{\Delta_{\mathcal{J}_c}} : \mathbf{m}\in\mathbb{R}^p \mapsto \sum_{i=1}^{p} m_i g_i(\mathbf{m}), \quad \mathbf{m} = \phi(\mathbf{M}_c(\cdot))$$
$$\text{with } g_i(\mathbf{m}) = \Delta_{\mathcal{J}_c}(\{\pi_1,\ldots,\pi_i\}) - \Delta_{\mathcal{J}_c}(\{\pi_1,\ldots,\pi_{i-1}\}) \quad (19)$$

where ϕ is softmax operation, and π being a permutation ordering of $\mathbf{m}$, such that $x_{\pi_1} \geq x_{\pi_2} \cdots \geq x_{\pi_p}$.

c) Adversarial loss: The primary objective of this loss is to help G_θ to model the underlying distribution of the true labels. The adversarial loss is defined as

$$\mathscr{L}_{\text{ADV}} = \mathbb{E}_{\mathbf{x}\sim\mathbb{P}_X,\hat{\mathbf{y}}\sim\mathbb{P}_{\mathbf{G}_\theta}} - \log(D_\psi(G_\theta(\mathbf{x},\hat{\mathbf{y}})). \quad (20)$$

Along with the mentioned losses, we have comprised the generator objective function by regularizing it with $L2$ regression with respect to the parameters of G to avoid overfitting. Therefore, the final objective for training the generator can be defined in the following equation:

$$\min_{\theta} \lambda_{\text{CE}}\mathscr{L}_{\text{CE}} + \lambda_{\text{IoU}}\mathscr{L}_{\mathcal{J}} + \lambda_{\text{ADV}}\mathscr{L}_{\text{ADV}} + \lambda_{L2}.||\theta||_2^2 \quad (21)$$

where λ_{CE}, λ_{IoU}, λ_{ADV}, and λ_{L2} represent the weights assigned to Focal loss, Lovász-Softmax Loss, adversarial loss, and L2 regularization, respectively.

IV. Experiments

In this section, we discuss experiments to analyze the efficacy of our proposed model for cloud segmentation.

A. Datasets and Study Area

We conducted our experiments on four different datasets. Primarily, we use publicly available the Biome dataset [26] consisting of 96 Landsat-8 scenes with different signatures (shown in Fig. 4) and 11 spectral bands with effective ground sampling distance (GSD) of 30 m to train our model. To visualize the effect of different bands in generated cloud masks, we have carried out experiments with different band combinations. Therefore, we have generated a paired dataset consisting of 70 000 ground truths and input patches with size $(256, 256)$ and $(256, 256, C)$, respectively. Here, $C \in \{1, 3, 4, 11\}$ corresponds to different band combinations of G, RGB, RGB + NIR, and all spectral bands, respectively. We use 30 000 samples for training, 20 000 for validation, and 20 000 for testing. Apart from this, we cross-validated our model on other datasets like SPARCS [27] consisting of 80 paired Landsat-8 patches of size (1000, 1000), Sentinel-2 (GSD = 10 m) cloud mask catalog [28] with 520 patches of size (1022 1022) and our self-crafted dataset from 50 scenes of the high-resolution multispectral sensor of Cartosat-2S satellite (GSD = 1.6 m). In the last two cases, we perform transfer learning with 2000 patches of shape (256 256), and due to the

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TABLE I

QUANTITATIVE ANALYSIS ON THAT BIOME DATASET AND COMPARISON WITH OTHER STATE-OF-THE-ART METHODS. FIRST AND SECOND METHODS ARE HIGHLIGHTED IN RED AND GREEN, RESPECTIVELY

Band Configuration	$C=1$							$C=3$						
Method	OA↑	Rec↑	Prec↑	mIoU(%)↑	Kappa↑	BI(%)↑	HD (m)↓	OA ↑	Rec↑	Prec↑	mIoU(%)↑	Kappa↑	BI(%)↑	HD (m)↓
CDNetV2 [10]	92.95	90.31	91.95	89.03	73.76	36.35	1911.76	91.28	87.67	90.38	86.86	77.29	32.49	1926.86
CDNetV1 [9]	88.79	88.31	86.78	84.83	73.65	36.81	1980.17	94.46	94.27	92.17	91.64	86.77	40.19	1508.82
DeepLabV3+ [6]	91.54	90.58	90.13	88.29	80.12	39.70	1683.65	91.82	91.43	90.21	88.61	81.17	39.92	1511.43
PSPnet [5]	92.99	92.55	90.67	89.68	83.08	39.89	1727.19	94.47	94.19	92.35	91.06	86.16	39.36	1556.73
RS-Net [11]	90.21	83.66	94.70	87.28	76.99	33.01	1723.82	93.53	92.42	94.46	91.09	84.36	36.16	1508.07
RefUNetV4 [12]	94.44	*94.46*	92.02	91.59	85.84	29.24	1365.91	*95.83*	**95.09**	94.33	*93.40*	*89.63*	*46.64*	*865.48*
ITF [17]	93.13	89.46	92.24	89.84	82.71	38.14	1643.92	93.42	90.54	91.35	90.67	84.33	38.89	1253.76
STCCD [16]	*94.64*	90.41	*94.95*	*91.68*	*86.88*	*42.23*	*1104.79*	94.95	*94.45*	*94.75*	92.66	88.94	43.71	952.48
C-LiSA (ours)	**97.02**	**96.31**	**95.80**	**94.77**	**91.88**	**55.19**	**601.19**	**97.52**	89.33	**96.43**	**95.29**	**92.82**	**56.90**	**577.42**
Band Configuration	$C=4$							$C=11$						
Method	OA↑	Rec↑	Prec↑	mIoU(%)↑	Kappa↑	BI(%)↑	HD (m)↓	OA ↑	Rec↑	Prec↑	mIoU(%)↑	Kappa↑	BI(%)↑	HD (m)↓
CDNetV2 [10]	89.52	91.56	86.12	84.56	75.92	34.68	1961.52	87.12	77.15	92.35	82.01	62.32	31.20	2637.38
CDNetV1 [9]	94.38	92.65	93.62	91.64	86.62	40.99	1538.30	94.34	92.68	93.71	91.47	86.04	40.47	1568.82
DeepLabV3+ [6]	93.48	93.23	92.07	90.57	85.03	39.29	1465.99	93.47	92.14	92.50	90.42	84.75	35.26	1583.98
PSPnet [5]	94.33	94.74	92.13	91.39	87.27	39.4	1381.27	95.11	*94.08*	93.93	92.30	88.12	41.21	1344.44
RS-Net [11]	92.11	90.58	91.66	89.30	84.05	42.58	1321.87	93.81	91.58	95.10	91.24	88.20	44.91	1178.96
RefUNetV4 [12]	95.36	93.48	94.67	93.07	88.04	47.81	890.69	95.33	94.00	93.99	93.98	87.52	47.03	941.49
ITF [17]	94.75	90.54	94.37	92.08	88.02	40.21	1130.52	95.05	92.77	94.64	92.49	89.08	45.75	1054.31
STCCD [16]	*95.84*	*94.07*	*94.89*	*93.74*	*90.01*	*50.19*	*735.34*	*96.16*	94.03	*96.08*	*94.32*	*91.23*	*51.55*	*630.18*
C-LiSA (ours)	**97.43**	**97.26**	**95.85**	**95.17**	**92.81**	**57.71**	**553.15**	**98.05**	**97.78**	**96.88**	**96.02**	**94.09**	**59.33**	**497.12**

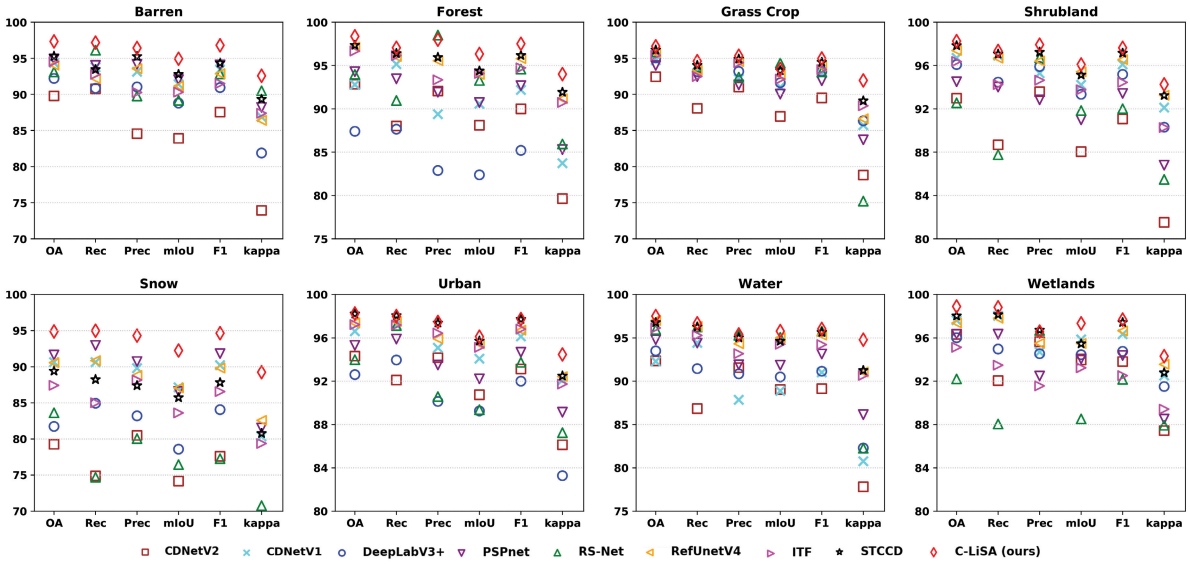


Fig. 4. Quantitative comparison on cloud mask generation with $C = 3$ in the presence of various signatures. C-LiSA has the most optimal performance in generating accurate and precise cloud masks irrespective of any complex signatures like snow.

difference in sensor definition, we only consider $C \in \{1, 3, 4\}$ band combinations for both.

B. Implementation Details

Here, we implement our G to extract features from five levels of deep layers due to our input shape of 256×256. This choice is the most optimal as this leads to the extraction of sufficient dense information while maintaining effective model size and complexity. Adding one more will increase the computation and parameters by 30% with minor changes in the features. We use an ADAM optimizer with a learning rate of 0.0001. During adversarial training, we update D once every single update in G. We set $\lambda_{\text{CE}} = 10$, $\lambda_{\text{IoU}} = 0.8$, $\lambda_{\text{ADV}} = 0.1$, and $\lambda_{L2} = 0.01$. Among other hyperparameters, we set the batch size as 64, γ in focal loss as 2.0 from the literature, and the maximum training iterations as 100. The entire framework is developed using PyTorch. All the experiments and training models are conducted on 2× Nvidia V100 with 32 GB GPU memory based on respective guidelines.

V. RESULTS AND ANALYSIS

In this section, we analyze the qualitative and quantitative performance of our proposed model. We compare it with other state-of-the-art methods in cloud segmentation, such as InTransformer (ITF) [17], STCCD [16], CDNetV2 [10], RS-Net [11], and RefUNetV4 [12]. In addition, we also use two representative segmentation modules, namely PSPnet [5] and DeeplabV3+ [6] for our comparative study. Along with traditional performance metrics like OA, recall (Rec), precision (Prec), and mean IoU (mIoU), we leverage the usage of kappa coefficient (Kappa), boundary IoU (BI) [29], and hausdraff distance (HD) for comprehensive analysis of our model performance.

A. Analysis on the Biome Dataset

1) Quantitative Analysis: Table I presents a quantitative comparison of our cloud mask generation model with other benchmark techniques on the Biome dataset. As shown in the table, our model outperforms others in terms of the

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TABLE II

QUANTITATIVE ANALYSIS ON THE SPARCS DATASET AND COMPARISON WITH OTHER STATE-OF-THE-ART METHODS. FIRST AND SECOND METHODS ARE HIGHLIGHTED IN RED AND GREEN, RESPECTIVELY

Band Configuration	$C = 1$							$C = 3$						
Method	OA↑	Rec↑	Prec↑	mIoU(%)↑	Kappa↑	BI(%)↑	HD (m)↓	OA↑	Rec↑	Prec↑	mIoU(%)↑	Kappa↑	BI(%)↑	HD (m)↓
CDNetV2 [10]	90.88	81.77	87.56	84.52	68.53	23.18	4732.56	92.07	82.83	90.65	86.07	72.12	26.62	3375.12
CDNetV1 [9]	90.21	80.43	86.49	83.83	66.02	25.76	5009.07	90.15	*85.84*	83.74	84.29	69.46	30.63	4233.48
DeepLabV3+ [6]	87.79	77.10	81.57	80.09	58.10	22.40	5546.19	85.88	79.61	77.29	78.86	56.72	28.04	4959.69
PSPnet [5]	82.62	75.9	72.73	75.28	48.22	18.92	4832.57	90.99	81.47	88.14	84.93	68.58	26.69	4173.38
RS-Net [11]	81.65	73.16	72.18	73.57	40.39	20.68	6758.24	83.73	70.09	74.18	74.68	43.65	23.85	4219.27
RefUNetV4 [12]	*91.65*	82.24	**92.83**	*86.22*	*72.29*	33.46	3965.76	92.03	83.37	*91.42*	86.84	73.37	34.08	3526.44
ITF [17]	88.39	80.17	82.41	81.16	68.24	35.08	3754.59	87.65	78.19	82.33	77.27	69.48	34.17	3612.67
STCCD [16]	90.68	*83.47*	85.48	84.16	70.27	*40.44*	*3017.83*	*92.14*	85.29	89.19	*88.44*	*74.38*	*43.72*	*2853.27*
C-LiSA (ours)	**93.85**	**86.09**	*90.56*	**89.13**	**73.53**	**58.26**	**1945.53**	**95.52**	**86.14**	**94.79**	**91.48**	**78.55**	**60.27**	**2182.82**
Band Configuration	$C = 4$							$C = 11$						
Method	OA↑	Rec↑	Prec↑	mIoU(%)↑	Kappa↑	BI(%)↑	HD (m)↓	OA↑	Rec↑	Prec↑	mIoU(%)↑	Kappa↑	BI(%)↑	HD (m)↓
CDNetV2 [10]	89.51	86.74	82.42	82.89	68.64	23.27	4248.81	88.42	70.35	92.95	80.78	52.34	18.72	8187.98
CDNetV1 [9]	91.88	83.95	88.95	86.10	72.32	31.03	4581.81	92.77	85.11	90.85	87.28	75.23	29.12	3856.01
DeepLabV3+ [6]	89.30	85.09	82.33	82.72	67.21	30.70	4549.95	92.58	84.71	90.57	86.95	74.52	28.20	3440.02
PSPnet [5]	*93.19*	*85.73*	**91.68**	86.83	76.03	31.36	3593.78	92.94	84.44	92.17	87.59	75.30	31.13	3269.02
RS-Net [11]	85.24	71.92	77.24	76.58	48.23	27.42	3844.34	85.41	70.35	78.38	76.58	46.67	27.79	4029.91
RefUNetV4 [12]	92.24	83.55	90.54	86.67	73.14	32.89	3184.41	93.26	84.42	**93.42**	88.16	76.15	35.58	3464.93
ITF [17]	92.08	88.67	82.49	86.09	72.26	39.63	2978.51	92.18	87.08	84.49	87.28	75.58	42.24	2654.19
STCCD [16]	92.62	84.14	90.68	*87.05*	**76.96**	*48.29*	*2615.27*	*94.69*	*87.14*	92.26	*90.08*	*79.57*	*52.88*	*2084.49*
C-LiSA (ours)	**94.06**	**87.61**	*91.51*	**89.08**	*76.62*	**62.64**	**1901.49**	**96.06**	**89.40**	*93.02*	**91.74**	**81.22**	**67.24**	**1713.19**

evaluation metrics for all band combinations. Compared to the second best method, our model not only provides 2%–3% improvement in OA and mIoU but also achieves significant performance gain of more than 3%–5% for Kappa, 7%–13% in BI, and 20%–45% in HD. This means our model can generate an accurate cloud mask with higher confidence and fewer spatial dissimilarities in the position and shape of segmented regions irrespective of the presence of any signatures. In Fig. 4, we visualize this phenomenon by comparing the performance of each model in the presence of different signatures with the most practical band-combination $C = 3$. From the figure, it is very apparent that our model has the most robust and favorable performance with respect to different metrics, especially in complex scenarios like the presence of snow, urban areas, and water bodies. From Table I, we also notice that other models choose to operate well with particular multiband combinations ($C = \{3, 4, 11\}$), whereas our model, C-LiSA works robustly for all band combinations resulting in better cloud mask prediction even with $C = 1$ due to the presence of multilevel cross-attention along with spatial and channel self-attention.

2) Qualitative Analysis: Fig. 5 shows cloud segmentation visual outcomes on the Biome dataset with different signatures. For effective visual assessment, we denote the cloudy and noncloudy pixels in white and black as well as mark the misinterpreted cloudy and noncloudy ones in red and cyan, respectively. As shown in the figure, our method generates cloud masks closest to GT compared to baseline techniques, especially for the signatures of urban, snow, and water bodies. Although not designed for remote sensing applications, PSPnet [5], and DeeplabV3+ [6] generate decent cloud masks in most cases, except in situations like thin cloud and snow-cloud coexistence. RefUNetV4 [12] can generate a cloud mask close to GT, however, it still suffers from omission and commission errors in scenarios like snow and urban regions. CDNetV2 [9], on the other hand, failed miserably in these situations by giving misclassified results most of the time as shown in the figure. ITF [17] and RS-Net [11] also suffer from similar errors resulting in low Kappa in Table I indicating that their performance is situational. Compared to them, STCCD [16], even in complex scenarios like snow-cloud coexistence, can generate accurate cloud masks to a certain extent and has low mispredictions. However, it still suffers from poor boundary accuracy despite using a boundary refinement module. In contrast, our method performs preferably by generating accurate cloud masks with precise boundary alignment with respect to the GT in most of the cases along with the said critical cases which also supports our quantitative analysis in Table I.

B. Analysis on SPARCS and Other Datasets

1) Quantitative Analysis: Table II illustrates quantitative results on the SPARCS dataset. Similar to the previous case, our method results in the best comprehensive outcomes for all band combinations. Compared to the second best method, our model achieves 1%–3% improvement in OA, 2%–3% in mIoU, 1%–4% in Kappa, 14%–18% in BI, and 17%–35% in HD. This observation is crucial as these results are generated using the weights corresponding to the training of the Biome dataset. Hence, this represents the generalization capability for each module. For better visualization, we have further estimated the percentage cloud coverage for each model and evaluated their accuracy with respect to the GT as shown in Fig. 6. From the figure, our estimated cloud coverage results are more closely distributed to the benchmark line for all 4 band combinations compared to others. This also can be depicted from the R2-scores and mean absolute error (MAE) values recorded in Fig. 6. Our method achieves the highest $R2$-score and lowest MAE for all band combinations ($C = \{1, 3, 4, 11\}$) followed by RefUNetV4 and STCCD. Other methods perform well for certain band combinations as shown in the scatter plots. This demonstrates that our method is more robust for any band configuration and has better generalization capability compared to other methods.

Table III refers to our quantitative study of cloud mask generation for Sentinel-2 and Cartosat-2S datasets using $C =$

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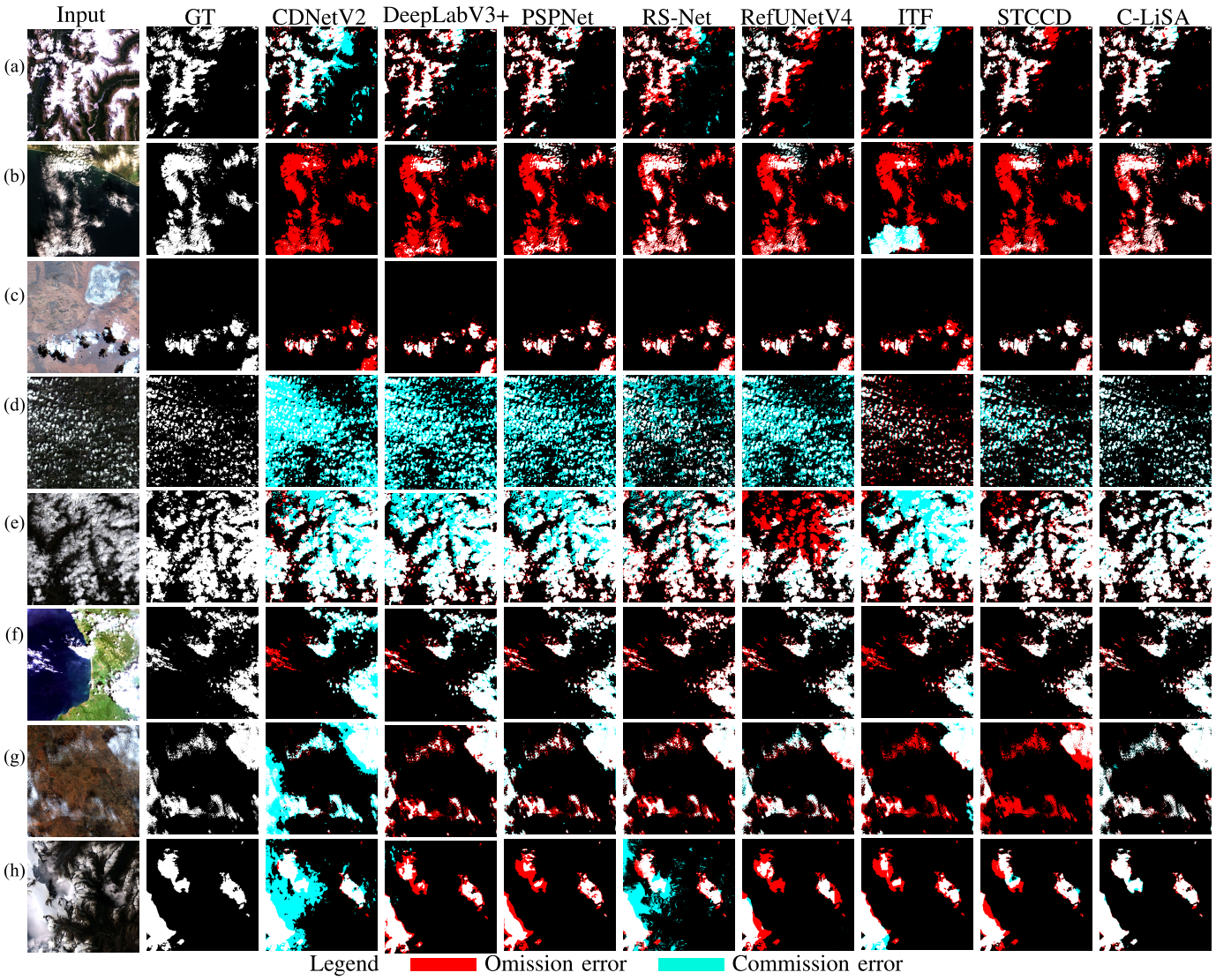


Fig. 5. Comparative cloud mask prediction on Biome dataset with different signatures. (a), (e), (h): snow, (b), (f): water-body, (c): dessert, (d): urban, and (g): barren. For each model the best output corresponding to the band combination (C) is shown. Irrespective of signatures, C-LiSA results in the most accurate cloud mask prediction with very precise boundaries.

TABLE III

QUANTITATIVE ANALYSIS ON THE SENTINEL-2 AND CARTOSAT-2S DATASETS AND COMPARISON WITH OTHER STATE-OF-THE-ART METHODS. FIRST AND SECOND METHODS ARE HIGHLIGHTED IN RED AND GREEN, RESPECTIVELY

Dataset	Sentinel-2 dataset																						
Band Configuration	$C=1$							$C=3$							$C=4$								
Method	OA↑	Rec↑	Prec↑	mIoU(%)↑	Kappa↑	BI(%)↑	HD (m)↓	OA↑	Rec↑	Prec↑	mIoU(%)↑	Kappa↑	BI(%)↑	HD (m)↓	OA↑	Rec↑	Prec↑	mIoU(%)↑	Kappa↑	BI(%)↑	HD (m)↓		
CDNetV2 [10]	70.87	70.63	74.96	65.82	40.16	22.59	2610.67	75.88	80.33	72.73	72.99	62.15	28.75	1139.44	72.96	72.71	74.13	70.98	60.77	27.15	1076.77		
CDNetV1 [9]	68.07	70.49	67.61	61.94	38.59	20.76	2680.75	72.42	72.29	72.75	65.75	50.46	23.90	1421.63	75.17	75.09	75.28	69.47	65.43	30.63	940.88		
DeepLabV3+ [6]	*76.13*	75.82	**77.81**	*70.34*	*60.05*	30.13	1142.08	82.27	82.43	82.67	77.19	72.61	36.31	579.37	81.03	82.17	80.36	78.82	74.11	35.98	493.61		
PSPnet [5]	66.13	65.93	60.73	54.64	44.11	18.54	3213.05	67.05	62.81	67.30	56.97	49.97	21.30	2149.90	69.05	67.67	68.44	60.80	54.96	21.11	1449.05		
RS-Net [11]	71.24	*76.34*	69.46	65.41	51.26	34.24	1002.81	83.96	84.81	*85.35*	80.15	71.83	39.27	509.64	81.68	81.43	*82.68*	75.75	70.19	40.34	559.58		
RefUNetV4 [12]	68.45	72.07	67.96	62.83	42.45	28.25	1133.11	73.9	74.72	75.56	71.78	61.51	32.51	884.18	78.93	78.80	79.35	73.56	66.81	35.53	763.10		
ITF [17]	63.11	62.25	59.46	55.06	48.49	24.55	1945.29	69.43	70.12	65.51	60.86	54.19	26.24	1780.76	70.26	71.36	67.28	63.11	56.47	28.29	1202.54		
STCCD [16]	72.29	74.48	69.13	64.25	59.84	*37.06*	*989.58*	*86.57*	*87.13*	83.68	*82.44*	*74.29*	*42.78*	*464.42*	*84.22*	*85.49*	81.35	*80.65*	*74.41*	*40.94*	*541.29*		
C-LiSA (ours)	**81.78**	**82.31**	*76.24*	**78.35**	**66.16**	**42.02**	**872.88**	**89.30**	**88.07**	**88.31**	**86.26**	**79.77**	**45.51**	**409.03**	**88.42**	**88.33**	**89.60**	**85.83**	**80.11**	**44.62**	**453.42**		
Dataset	Cartosat-2S																						
Band Configuration	$C=1$							$C=3$							$C=4$								
Method	OA↑	Rec↑	Prec↑	mIoU(%)↑	Kappa↑	BI(%)↑	HD (m)↓	OA↑	Rec↑	Prec↑	mIoU(%)↑	Kappa↑	BI(%)↑	HD (m)↓	OA↑	Rec↑	Prec↑	mIoU(%)↑	Kappa↑	BI(%)↑	HD (m)↓		
CDNetV2 [10]	89.27	88.69	87.87	85.61	79.29	48.46	135.81	90.59	91.40	90.02	85.46	80.15	50.29	112.34	87.51	89.48	85.27	82.54	75.16	47.59	104.78		
CDNetV1 [9]	89.98	89.04	88.68	83.34	75.48	46.78	125.44	91.16	91.07	91.64	84.78	78.66	52.78	100.49	87.08	87.26	87.78	84.04	74.69	48.12	96.42		
DeepLabV3+ [6]	85.49	88.12	82.77	80.94	72.65	44.49	150.22	92.86	92.22	92.68	88.59	85.41	50.05	80.32	89.38	88.14	88.41	88.06	79.66	52.73	88.90		
PSPnet [5]	92.44	93.08	93.26	88.42	81.49	50.49	96.36	87.88	89.92	88.40	80.19	76.64	43.34	124.44	85.84	86.62	84.18	79.26	72.76	45.57	112.33		
RS-Net [11]	93.78	91.49	*94.07*	91.08	86.49	54.24	82.71	93.81	91.54	*95.78*	90.15	84.86	62.27	55.91	91.22	90.04	90.23	89.48	82.24	56.48	99.74		
RefUNetV4 [12]	94.29	**96.08**	92.00	91.46	85.78	60.46	74.95	96.03	95.27	95.49	93.46	90.49	69.57	36.00	91.47	90.80	91.78	89.24	87.46	61.25	84.76		
ITF [17]	93.17	94.02	89.48	91.14	85.04	51.56	90.42	94.51	94.78	90.12	91.26	85.59	60.22	59.41	90.27	89.84	85.51	85.92	79.04	53.48	96.57		
STCCD [16]	*94.77*	93.48	92.14	*92.39*	*87.56*	*66.48*	*57.67*	*96.62*	*95.76*	94.24	*94.59*	*91.74*	*72.41*	*30.38*	*92.66*	*91.14*	*93.49*	*90.24*	*87.93*	*63.27*	*68.44*		
C-LiSA (ours)	**96.82**	*95.40*	**94.74**	**95.22**	**93.25**	**75.51**	**28.81**	**97.90**	**96.35**	**96.16**	**96.33**	**94.76**	**78.54**	**18.89**	**95.28**	**92.33**	**96.58**	**94.26**	**90.11**	**72.56**	**41.69**		

{1, 3, 4} with transfer learning for two epochs. For Sentinel-2, our proposed method achieves a performance gain of 3%–5% in OA, 4%–8% in mIoU, 5%–6% in Kappa, 3%–5% in BI, and 12%–16% in HD, whereas for Cartosat-2S, we observe a 1%–3% improvement in OA, 2%–4% in mIoU, 3%–6% in Kappa, 6%–10% in BI, and 40%–50% in HD with

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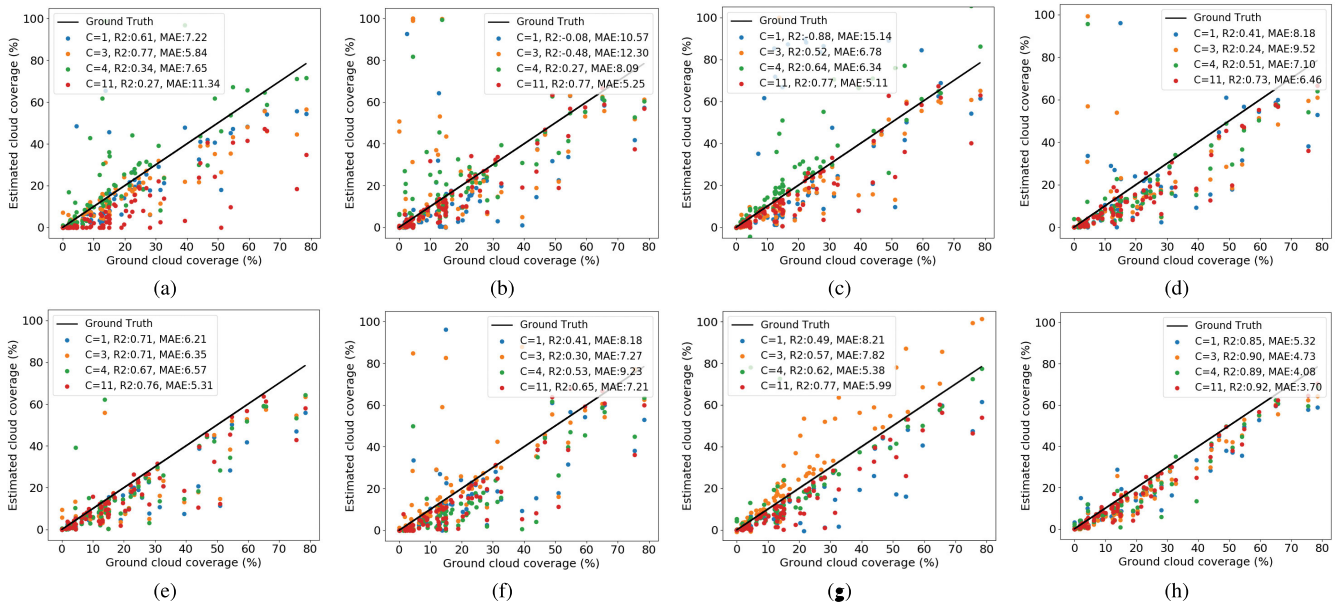


Fig. 6. Comparison cloud coverage estimation on SPARCS dataset among (a) CDNetv2. (b) DeepLabV3+. (c) PSPnet. (d) RS-Net. (e) RefUNetV4. (f) ITF. (g) STCCD. (h) C-LiSA (ours).

respect to corresponding second best scenario. Therefore, for cross-domain cloud mask generation, our model has the best generalization potential and can perform robustly with minimal training requirements.

2) Qualitative Analysis: Fig. 7 illustrates visual results of cloud mask generation for three datasets, SPARCS, Sentinel-2, and Cartosat-2S. Similar to the Biome scenario, our proposed method outperforms all other benchmark methods for all the datasets. CDNetV2 in all aspects is unable to perform effectively, especially for complex scenarios like snow-cloud coexistence and thin clouds. On the other hand, ITF and DeepLabV3+ can distinguish between snow and cloud features for the SPARSAC dataset. However, they are unable to perform well for the other two datasets and suffer from omission errors. The performance of PSPnet and RS-Net is situational, and for most of the complicated cases, they also result in misclassified outputs. RefUNetV4 and STCCD perform well in these cases but exhibit occasional omission errors and limited boundary accuracy. Comparatively, our proposed model can precisely extract cloud masks with minimal error in all situations and also follows the boundary of the segmented masks very closely with respect to GT. This also supports our quantitative analysis in the above tables.

VI. Ablation Study

In this section, we perform ablation studies to support our design and configuration choices.

A. Choice of Objective Functions

Table IV shows quantitative results for the effect of each loss function on the performance of our model on the Biome dataset with $C = 11$ band configuration. As discussed in Section III-B, introducing Lovász-Softmax Loss instead of traditional Jaccard loss improves the overall performance by 2% in OA, more than 6% in mIoU and 10% in BI. These results are further enhanced by introducing adversarial loss, which adds a performance gain of 4% in Kappa and 27% in HD. Fig. 8 visualizes the effect of each loss on the estimated cloud mask. We can see that due to the introduction of Lovász loss, our model can generate a cloud mask with better boundary accuracy and fewer misinterpreted cases. In addition, the inclusion of adversarial loss helps our model to estimate cloud masks aligned to reference masks.

TABLE IV

Quantitative Ablation Study to Study Contribution of Different Applied Loss Functions

Focal+L2 Loss	Jaccard Loss	Lovasz Loss	Adversarial Loss	OA↑	Rec↑	Prec↑	mIoU(%)↑	Kappa↑	BI(%)↑	HD (m)↓
✓	✗	✗	✗	93.97	93.58	94.35	86.88	88.49	42.29	923.57
✓	✓	✗	✗	95.29	93.98	93.76	88.06	89.06	45.59	952.18
✓	✗	✓	✗	97.12	96.44	94.73	94.25	90.26	55.71	682.15
✓	✗	✓	✓	**98.05**	**97.78**	**96.88**	**96.02**	**94.09**	**59.33**	**497.12**

TABLE V

Quantitative Ablation Study for Introducing Different Proposed Architectural Elements

DOSA	DOSA+HC2A	OA↑	Rec↑	Prec↑	mIoU(%)↑	Kappa↑	BI(%)↑	HD (m)↓
✗	✗	94.09	96.41	92.88	92.25	90.11	48.45	742.29
✓	✗	95.26	94.87	94.02	93.73	92.19	53.37	662.28
✓	✓	**98.05**	**97.78**	**96.88**	**96.02**	**94.09**	**59.33**	**497.12**

B. Choice of Architectural Components

We study the significance of two attention modules, DOSA, and HC2A in generating precise cloud masks. Table V illustrates the comprehensive performance of our model on the Biome dataset for $C = 11$ in the presence of proposed attention modules. From the table, it is clear that introducing DOSA and HC2A improves in performance of our model by 4% in OA, mIoU, and Kappa, 11% in BI, and 33% in HD with respect to the baseline model. Fig. 9 shows the effect

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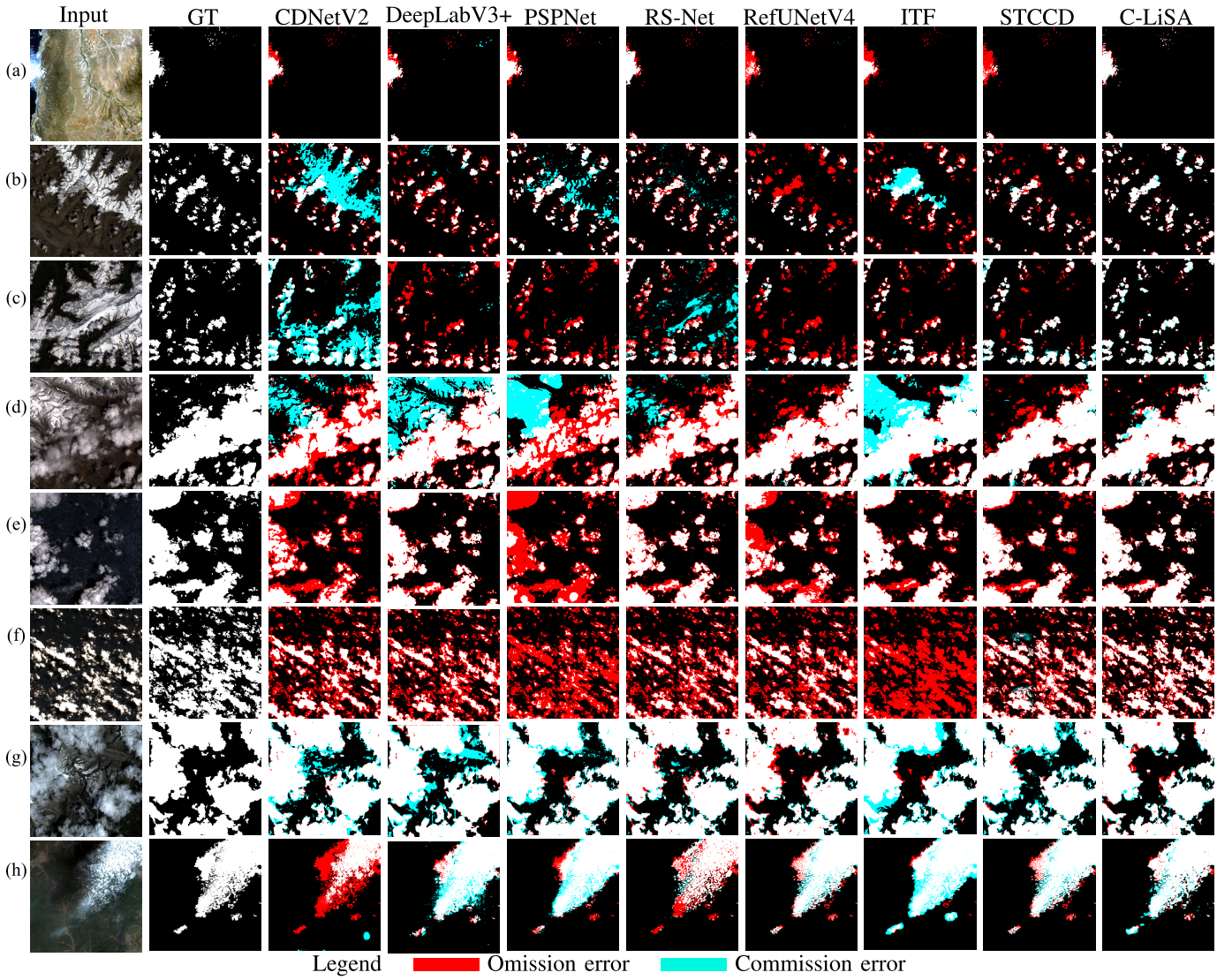


Fig. 7. Comparison of cloud mask generation results on different datasets. (a)–(c) SPARCS dataset. (d)–(f) Sentinel-2 dataset. (g) and (h) Cartosat-2s dataset. In all the datasets, C-LiSA results in the most accurate cloud mask which suggests its favorable generalization capability.

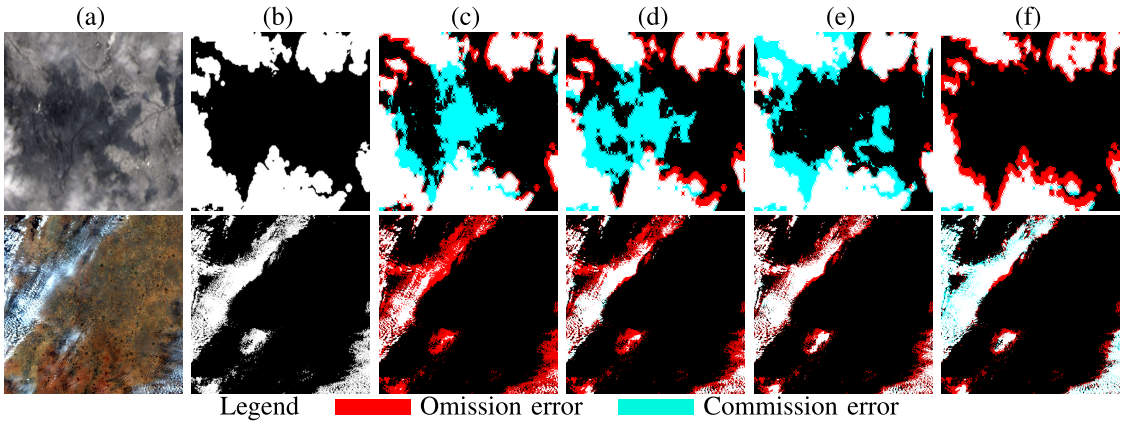


Fig. 8. Ablation study for adding different loss functions. (a) Input. (b) GT. (c) Focal loss. (d) Focal + Jacard loss. (e) Focal + Lovasz loss. (f) Focal + Lovasz + adversarial loss. Clearly, utilization of Lovasz loss in an adversarial setup gives close to GT cloud mask with very precise boundary accuracy.

of these attention modules in cloud segmentation visually. Without attention modules, the model results in commission errors. The introduction of DOSA mitigates this issue up to a certain extent, however, it also increases omission error as depicted in the figure. Presenting HC2A along with DOSA reduces the overall misclassifications. It can be visualized from

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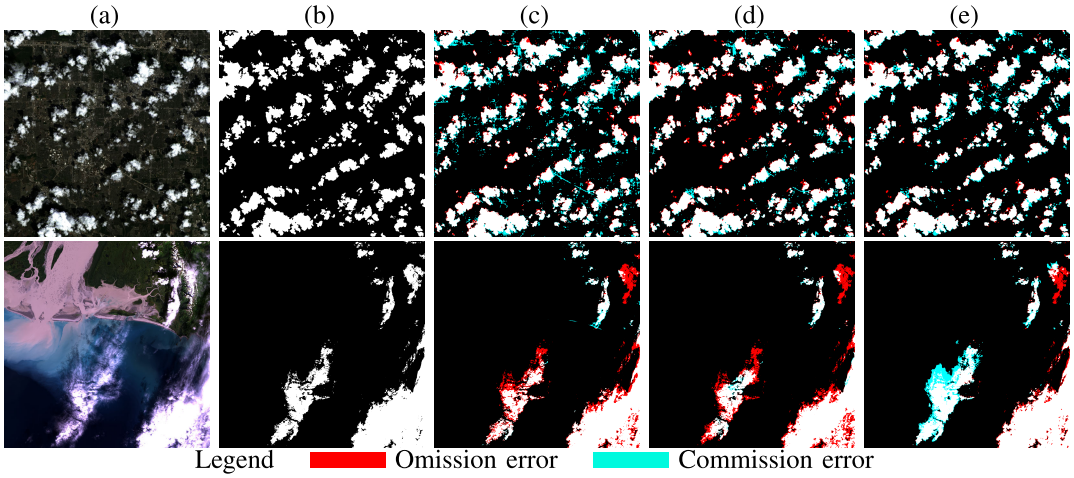


Fig. 9. Ablation study related to different proposed architectural elements. (a) Input. (b) GT. (c) Without attention. (d) DOSA. (e) DOSA + HC2A. Clearly, introducing both attention modules results in precise cloud mask extraction with accurate boundary alignment and better generalization.

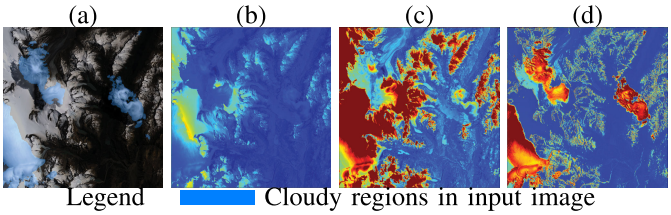


Fig. 10. Feature map visualization after each attention module. (a) Input with GT cloud mask indicated by, feature map. (b) Without attention (X_1). (c) After DOSA (Z_1). (d) After DOSA + HC2A (Y_1).

TABLE VI

SOME OF EMPIRICALLY ESTIMATED 2-LIPSCHITZ CONSTANT OF PROPOSED ATTENTION MODULES FOR DIFFERENT PERTURBED CHANNELS

i	$\|\|J_{i,i}\|\|_2$ for MST	$\|\|J_{i,i}\|\|_2$ for DOSA	$\|\|J_{i,i}\|\|_2$ for DOSA+HC2A
0	2.32	0.86	0.69
5	1.59	0.94	0.74
25	1.88	1.26	0.78
50	0.78	0.73	0.62

the second example of Fig. 9, where there is an omission error in reference to the positions of thin clouds. However, using both DOSA and HC2A, our model can faithfully mark these cases as cloudy pixels. The main reason behind these can be visualized in Fig. 10 where we present an illustration of the feature map after each attention module. It is readily understandable that without any attention, the model prioritizes every feature equally. Whereas applying only DOSA, focuses more on cloudy regions, however, it also focuses on redundant confusing features such as snow. After employing DOSA + HC2A, it not only gives more emphasis to the cloudy pixels but also suppresses the surrounding redundant features.

C. Lipschitz Stability and Adversarial Robustness

To validate the proposed Theorems 1 and 2, we empirically estimate the 2-Lipschitz constant by perturbing specific input channel (denoted as i in Table VI) of the feature map corresponding to encoder output at the first level when we pass Fig. 5(a) input image to our model. The diagonal elements of Jacobian corresponding to computing this Lipschitz coefficient are estimated by observing the related change in the output of the attention modules based on the perturbed input. Table VI shows the estimated Lipschitz constant for our DOSA + HC2A module and is also compared with that of multihead-self-attention transformer (MST). As shown in the table, the spectral norm of diagonal elements of the Jacobian corresponding to different perturbed channels is comparatively less for our proposed attention modules with respect to that of MST. This not only supports our argument from the derived Theorems but also suggests that our proposed method has better stability in terms of generating robust output in the presence of any perturbed input. Due to this enhanced Lipschitz stability, it also implies robustness against adversarial attacks as studied in [30]. To validate this, we perform experiments of our model against the fast gradient sign method (FGSM) and projected gradient descent (PGD) attacks and compare the performance with the attention module of MST and Restormer by employing them instead of DOSA + HC2A in our model. For all the cases, we perturb the input MX image by tuning step size for both FGSM and PGD-20 attacks. As shown in Fig. 11, our method results in more stable performance in terms of mIoU compared to other self-attention modules as the perturbation to the input increases for both attacks. Therefore, this supports our argument of providing better stability by introducing the DOSA + HC2A module which also provides better robustness against adversarial attacks.

VII. OVERALL COMPLEXITY ANALYSIS

Here, we analyze the complexity of our attention modules for an input feature X of shape $H \times W \times C$. For DOSA, the computation of attention scores in spatial and channel attention branches involves: 1) matrix multiplication between key and value arrays with a computational complexity of $\mathcal{O}(HWC)$ and 2) element-wise multiplication with value with complexity $\mathcal{O}(HWC)$ resulting in overall $\mathcal{O}(HWC)$ complexity, making it a linear self-attention. The estimation of attention in HC2A has complexity $\mathcal{O}(C^2)$, followed by matrix multiplication with

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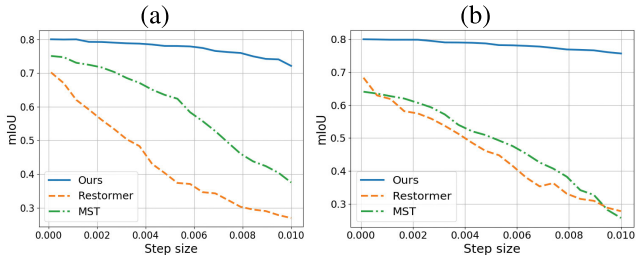


Fig. 11. Analysis of adversarial robustness against (a) FGSM and (b) PGD-20, by using attention of Restormer and MST instead of C-LiSA in our model.

TABLE VII
MODEL SIZE AND COMPUTATIONAL COMPLEXITY ANALYSIS

Methods	GFLOPs (1G = 1×10^9)	Trainable params (1M = 1×10^6)	Inference Time (s)
CDNetV2 [10]	31.5	65.9 M	1.93
CDNetV1 [9]	48.5	64.8 M	1.98
DeepLabV3+ [6]	31.8	40.3 M	1.96
PSPnet [5]	19.3	46.6 M	1.81
RS-Net [11]	28.2	37.8 M	1.78
RefUNetV4 [12]	30.2	41.3 M	1.86
ITF [17]	167.6	109.8 M	2.15
STCCD [16]	69.5	48.3 M	1.97
C-LiSA (ours)	73.9	44.6 M	1.89

value of shape $HW \times C$. Therefore, the overall memory complexity of HC2A is $\mathcal{O}(HWC^2)$. Therefore, the total complexity for DOSA + HC2A is $\mathcal{O}(HWC + HWC^2)$.

In Table VII, we also investigate the computational complexity of our framework and compare it with other benchmark methods in terms of the number of floating-point operations (FLOPs), number of trainable parameters, and inference time. FLOPs are computed from [31] with an input image shape of (224, 224). Inference time is estimated for an input shape of (1024, 1024). It can be seen that our model has less number of parameters compared to other hybrid transformers resulting in competitive inference time. However, our model has a relatively higher number of FLOPs due to the operation of HC2A with high channel dimensions in the deeper layers. Still, compared to other hybrid transformer models, it is relatively close to that of [16] and 57% less than that of [17].

VIII. CONCLUSION

In this study, we propose a novel hybrid transformer architecture, namely the C-LiSA network. Here, we establish how Lipschitz stability is necessary for the robust performance of a model against different adversarial perturbations. In this regard, we develop two novel attention modules DOSA and hierarchical cross-channel attention (HC2A), and theoretically justify their effectiveness in providing better Lipschitz stability compared to other benchmark methods. By building our whole framework under an adversarial setting in the presence of Lovász-Softmax Loss and carrying out experiments on multiple satellite datasets like Landsat-8, Sentinel-2, and Cartosat-2S, we empirically show that due to Lipschitz stability, our model not only generates cloud mask more robustly under various situations like different band-combinations and different resolutions of satellite images but also results in accurate cloud mask with precise boundary alignment with respect to GT compared to other state-of-the-art methods. Hence, our proposed framework for cloud segmentation ensures better generalization as well as superior boundary accuracy of generated cloud masks. We also conducted an ablation study to verify these claims. In the future, we would like to improve the computational performance of our model for real-time mask estimation and also extend our work for multiclass segmentation in satellite images with the same precision and accuracy.

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Subhajit Paul received the B.Tech. degree in electronics and communication engineering from the Indian Institute of Space Science and Technology (IIST), Thiruvananthapuram, India, in 2020.

He joined the Signal and Image Processing Area (SIPA) in SAC, ISRO, Ahmedabad, India, in 2021, where he is currently working as a Scientist-SC. His research interests include different theoretical aspects in computer vision, optimization, and generative AI with applications related to terrestrial camera data as well as satellite data processing.



Ashutosh Gupta received the B.E. degree in computer science engineering from Government Engineering College, Jabalpur, India, in 2009, and the M.E. degree in system science and automation from IISc., Bengaluru, India, in 2016.

In 2009, he joined SAC, ISRO, Ahmedabad, India, where he is currently working as a Scientist-SF. His research interests include image processing, computer vision, artificial intelligence, blockchain technology, and high-performance computing for Earth observation and planetary missions of ISRO.



S. Manthira Moorthi received the Ph.D. degree in computational physics from SRM University, Chennai, India, in 2019.

He joined as a Scientist with SIPA, SAC, ISRO, Ahmedabad, India, in 1991. He has been contributing to the remote sensing data processing domain in the development of methods, algorithms, and software systems for pre- and postprocessing systems for Earth and planetary observations. His research interests include satellite image rectification, registration, restoration, and AI/ML-based image processing.



Debajyoti Dhar received the B.E. degree in computer science and technology from IIEST, Kolkata, India, in 1987, and the M.Sc. degree in computer science from BITS, Pilani, India, in 2013.

He is currently an Outstanding Scientist and the Deputy Director of the Signal and Image Processing Area (SIPA), SAC, ISRO, Ahmedabad, India, In a span of over 30 years with ISRO, he worked in system programming, device driver, Unix Kernel programming. His research interests include applied mathematics, computer vision, machine, and deep learning.

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